

So, suppose that $\forall x (P(x) \wedge Q(x))$ is true. This means that if a is in the domain, then $P(a) \wedge Q(a)$ is true. Hence, $P(a)$ is true and $Q(a)$ is true. Because $P(a)$ is true and $Q(a)$ is true for every element a in the domain, we can conclude that $\forall x P(x)$ and $\forall x Q(x)$ are both true. This means that $\boxed{\forall x P(x) \wedge \forall x Q(x)}$ is true.

Next, suppose that $\forall x P(x) \wedge \forall x Q(x)$ is true. It follows that $\forall x P(x)$ is true and $\forall x Q(x)$ is true. Hence, if a is in the domain, then $P(a)$ is true and $Q(a)$ is true [because $P(x)$ and $Q(x)$ are both true for all elements in the domain, there is no conflict using the same value of a here]. It follows that for all a , $P(a) \wedge Q(a)$ is true. It follows that $\forall x (P(x) \wedge Q(x))$ is true. We now conclude that $\boxed{\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)}$

1.4.7 Binding Variables

When a quantifier is used on the variable x , we say that this occurrence of the variable is bound. An occurrence of a variable that is not bound by a quantifier or set equal to a particular value is said to be free.

The part of a logical expression to which a quantifier is applied is called the scope of this quantifier. Consequently, a variable is free if it is outside the scope of all quantifiers in the formula that specify this variable.

Example 18: In the statement $\exists x (x+y=1)$, the variable x is bound by the existential quantification $\exists x$, but the variable y is free because it is not bound by a quantifier and no value is assigned to this variable. This illustrates that in the statement $\exists x (x+y=1)$, x is bound, but y is free.

In the statement $\boxed{\exists x (P(x) \wedge Q(x)) \vee \forall x R(x)}$, all variables are bound. The scope of the first quantifier, $\exists x$, is the expression $P(x) \wedge Q(x)$ because $\exists x$ is applied only to $P(x) \wedge Q(x)$ and not to the rest of the statement. Similarly, the scope of the second quantifier, $\forall x$, is the expression $R(x)$. That is, the existential quantifier binds the variable x in $P(x) \wedge Q(x)$ and the universal quantifier $\forall x$ binds the variable x in $R(x)$.